Conformal Inference Assignment 3
(Deadline: November 1)

Wen, Zehai

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Remark All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

Exercise 1

Given exchangeable data set $(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$, level $\alpha \in (0, 1)$, and a real valued symmetric score function $s((x, y); \mathcal{D})$ for $(x, y) \in \mathcal{X} \times \mathcal{Y}$ and $\mathcal{D} \in (\mathcal{X} \times \mathcal{Y})^k$ for any k being a data set, we carry out the full conformal prediction. Put, for any $y \in \mathcal{Y}$:

$$\mathcal{D}_{n+1}^y = \{(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, y)\}.$$

For simplicity, write $S_i^y = s((X_i, Y_i); \mathcal{D}_{n+1}^y)$ for $i \in \{1, \ldots, n\}$ and $S_{n+1}^y = s((X_{n+1}, y); \mathcal{D}_{n+1}^y)$. The standard conformal prediction procedure asks us to compare S_{n+1}^y with the entire sequence $(S_1^y, \ldots, S_n^y)$.

We proceed however by choosing randomly a subset $I \subseteq \{1, \ldots, n\}$ independent of the data and compare S_{n+1}^y with $(S_i^y)_{i \in I}$. More precisely, put:

$$\mathcal{C}(X_{n+1}) = \left\{ y \in \mathcal{Y} \mid S_{n+1}^y \leqslant \text{Quantile}\left((S_i^y)_{i \in I}; (1-\alpha)\left(1 + \frac{1}{|I|}\right)\right)\right\}.$$

Prove that

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \geqslant 1 - \alpha.$$

Solution Given that I is independent of the data set, the joint distribution of $((X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, Y_{n+1}))$ remains the same after conditioning on I. In particular, they are still exchangeable. Therefore:

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} = \sum P\{I = J\} P\left(\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} \mid I = J\right) \geqslant \sum P\{I = J\}(1 - \alpha) = 1 - \alpha,$$

where the summation is taken with J ranged over all subsets of $1, \ldots, n$. The second inequality is Theorem 3.2 in the textbook. ////

Exercise 2

Fix positive integer K and consider the K-nearest neighbor regression with $\mathcal{X}$ being any set (any metric space) and $\mathcal{Y} = \mathbb{R}$ who gives a fitted model

$$\hat{f}(x; \mathcal{D}) = \frac{1}{K} \sum_{i \in N(x, \mathcal{D})} y_i,$$

where $\mathcal{D}$ is any data set containing points from $\mathcal{X} \times \mathcal{Y}$ and $N(x, \mathcal{D})$ denotes indices of all K points in $\mathcal{D}$ whose $\mathcal{X}$ component is, in some sense, closest to x. We can assume that there is almost surely no ties by implementing any tie breaking rule.

Now let us consider the data set

$$\mathcal{D}_n = \{(X_1, Y_1), \cdots, (X_n, Y_n)\}.$$

For any $y \in \mathcal{Y}$, put

$$\mathcal{D}_{n+1}^y = \{(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, y)\}.$$

Prove that there exists real valued functions $a(x; \mathcal{D}_n, X_{n+1})$ and $b(x; \mathcal{D}_n, X_{n+1})$ whose values depend only on x, $\mathcal{D}_n$, and X_{n+1} such that, for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$:

$$\hat{f}(x; \mathcal{D}_{n+1}^y) = a(x; \mathcal{D}_n, X_{n+1}) + b(x; \mathcal{D}_n, X_{n+1}) y.$$

Solution First observe the fact that the set $N(x, \mathcal{D}_{n+1}^y)$ merely reads the set of all indices $i \in \{1, \ldots, n+1\}$ belonging to the K points amongst $X_1, \ldots, X_{n+1}$ who are the closest to x. Therefore, although y is present in the notation, $N(x, \mathcal{D}_{n+1}^y)$ depends only on x and the points $X_1, \ldots, X_{n+1}$. Therefore, by simply writing

$$\hat{f}(x; \mathcal{D}_{n+1}^y) = \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{1, \ldots, n\}} y_i + \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{n+1\}} y_i = \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{1, \ldots, n\}} y_i + \frac{1}{K} \chi_{\left\{n+1 \in N(x, \mathcal{D}_{n+1}^y)\right\}} y,$$

we see that it suffices to define

$$a(x; \mathcal{D}_n, X_{n+1}) = \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{1, \ldots, n\}} y_i$$

and

$$b(x; \mathcal{D}_n, X_{n+1}) = \frac{1}{K} \chi_{\left\{n+1 \in N(x, \mathcal{D}_{n+1}^y)\right\}}.$$

This completes the proof.////

Exercise 3

Given data set $(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$ with $\mathcal{Y} = \mathbb{R}$, level $\alpha \in (0, 1)$, we discretize $[-B, B]$, where $B > 0$, by setting, for a positive integer M and all $m \in \{1, \ldots, M\}$:

$$y^{(m)} = -B + \frac{m-1}{M-1} \times 2B.$$

Put, for any $y \in \mathcal{Y}$:

$$\mathcal{D}_{n+1}^{y} = \{(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, y)\}.$$

With a real valued score function $s((x, y); \mathcal{D})$ for $(x, y) \in \mathcal{X} \times \mathcal{Y}$ and $\mathcal{D}$ is any data set containing points from $\mathcal{X} \times \mathcal{Y}$, we consider a discretized conformal prediction set defined as follows. First of all, let I be the set of all $m \in \{1, \ldots, M\}$ such that

$$s\left((X_{n+1}, y^{(m)}); \mathcal{D}_{n+1}^{y^{(m)}}\right) \leqslant \text{Quantile}\left(\left\{s\left((X_i, Y_i); \mathcal{D}_{n+1}^{y^{(m)}}\right)\right\}_{i=1}^{n}; (1-\alpha)\left(1 + \frac{1}{n}\right)\right).$$

Then, put

$$\mathcal{C}(X_{n+1}) = \bigcup_{m \in I} (y^{(m-1)}, y^{(m+1)}).$$

Construct an example of a symmetric score function s and a distribution P of the data set such that

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} = 0$$

for any choice of B and M.

Solution We construct the example as follows. First of all, map the set $\mathcal{X}$ to $\mathbb{R}$ in any way one wants. From now on, we shall identify $\mathcal{X}$ with $\mathbb{R}$ and we will write for simplicity $\mathcal{X} = \mathbb{R}$. We construct it such that the pairs (X_i, Y_i) ranged over $i \in \{1, \ldots, n+1\}$ are independent identically distributed with distribution of X_i's being any fixed continuous probability measure (say, standard normal) and $Y_i = X_i$. As of the score, we simply put

$$s((x, y); \mathcal{D}) = |x - y|$$

for all (x, y) and all data set $\mathcal{D}$. Clearly, s is symmetric because it does not even depend on $\mathcal{D}$. If $i \in \{1, \ldots, n\}$, our construction yields, almost surely, that

$$s\left((X_i, Y_i); \mathcal{D}_{n+1}^{y^{(m)}}\right) = |X_i - Y_i| = 0$$

and

$$\text{Quantile}\left(\left\{s\left((X_i, Y_i); \mathcal{D}_{n+1}^{y^{(m)}}\right)\right\}_{i=1}^{n}; (1-\alpha)\left(1 + \frac{1}{n}\right)\right) = 0$$

Therefore, $I = \emptyset$ and $\mathcal{C}(X_{n+1}) = \emptyset$ for all choice of M and all choice of B. ////

Exercise 4

For a data set $\mathcal{D} = ((X_1, Y_1), \ldots, (X_m, Y_m)) \in (\mathcal{X} \times \mathcal{Y})^m$, where $\mathcal{Y} = \mathbb{R}$, and any permutation $\sigma \in S_m$, put

$$\mathcal{D}_\sigma = ((X_{\sigma(1)}, Y_{\sigma(1)}), \ldots, (X_{\sigma(m)}, Y_{\sigma(m)})).$$

Given model $\hat{f}(x; \mathcal{D})$, we construct a randomized score function $s((x, y); \mathcal{D}, \xi)$, where ξ is a random variable independent of the data set, as follows. Let σ_ξ be an S_m-valued random variable depending only on ξ that has the uniform distribution on S_m if ξ has the uniform distribution on $[0, 1]$. Put

$$s((x, y); \mathcal{D}, \xi) = \left| y - \hat{f}(x; \mathcal{D}_{\sigma_\xi}) \right|$$

for all (x, y) and all values of ξ. Prove that s is a symmetric score function in the following sense: for every fixed test points $(x_1^{'}, y_1^{'}), \cdots, (x_{m'}^{'}, y_{m'}^{'})$ and every $\tau \in S_m$, the vector

$$\left(s\left((x_1^{'}, y_1^{'}); \mathcal{D}, \xi\right), \ldots, s\left((x_{m'}^{'}, y_{m'}^{'}); \mathcal{D}, \xi\right) \right) \in \mathbb{R}^{m'}$$

has the same distribution as

$$\left(s\left((x_1^{'}, y_1^{'}); \mathcal{D}_\tau, \xi\right), \ldots, s\left((x_{m'}^{'}, y_{m'}^{'}); \mathcal{D}_\tau, \xi\right) \right) \in \mathbb{R}^{m'}$$

when ξ has the uniform distribution on $[0, 1]$.

Solution For each $i \in \{1, \ldots, m'\}$, the random variable

$$s\left((x_i^{'}, y_i^{'}); \mathcal{D}, \xi\right) = \left| y_i^{'} - \hat{f}(x_i^{'}; \mathcal{D}_{\sigma_\xi}) \right|$$

has the same distribution as

$$\left| y_i^{'} - \hat{f}(x_i^{'}; \mathcal{D}_{\tau \circ \sigma_\xi}) \right| = s\left((x_i^{'}, y_i^{'}); \mathcal{D}_\tau, \xi\right)$$

as long as ξ has the uniform distribution on $[0, 1]$ because, by construction, both σ_ξ and $\tau \circ \sigma_\xi$ have uniform distribution on S_m. ////